

Unpacking the Math of the Blinds: A Statistical Review of Luck vs. Skill in Poker

Is poker a game of skill, or just a high-stakes gambling game with luck? For decades, this question has breached casino walls and entered federal courtrooms, shaping the legal landscape of online gaming.

A fascinating detailed discussion of this topic can be found on Andrew Gelman's *Statistical Modeling, Causal Inference, and Social Science* blog, specifically in the post [Luck vs. Skill in Poker](#). The post centers around a compelling framework proposed by statistician Rick Schoenberg in his book *Probability with Texas Hold'em Applications*.

Well, just like Schoenberg suggested, "Luck is the equity gained or lost during the deal of the cards, while skill is the equity gained or lost during the betting rounds." What does that mean exactly? We can use a very simple example to demonstrate. First we need to define equity.

Mathematically speaking,

$$\text{Equity} = \text{current pot size} \times \text{Probability of winning at showdown}$$

Assuming no future betting occurs.

Schoenberg breaks every hand down chronologically. When the dealer slides the hole cards across the felt, or drops the flop, turn, or river, any shift in a player's probability of winning is considered as **luck**. On the other hand, when a player bets, raises, or folds, any change in their expected profit is categorized as **skill**.

Consider an example highlighted in the blog:

On one hand, Minieri is dealt A♣ J♣ and Lederer holds A♥ 9♥.

Luck portion: Pre-flop, Minieri's hand has a 70.07% chance of winning. Because the deck favored him, his "luck equity" immediately surges by several hundred chips.

Skill portion: Minieri leverages his advantage by putting in a pre-flop raise. Because he forces Lederer to contribute chips to the pot. Minieri extracts massive positive expected value (+EV). In Schoenberg's model, this calculation registers as a major spike in the **skill** column. For more examples, you can take a look at [*Probability with Texas Hold'em Applications*](#).

However, While this mathematical breakdown is clean, both Andrew Gelman and subsequent commenters highlight a gaping vulnerability in the framework: **luck frequently leaks into the skill metric**. Imagine a classic hand: Player A is dealt pocket Aces (AA) and Player B is dealt pocket Kings (KK). Pre-flop, the Aces have an 82% equity advantage. Naturally, both players will aggressively reraise until all the chips are in the middle. By Schoenberg's definition, because Player A is growing the size of the pot while holding the best hand, every chip added represents an accumulation of *skill* equity. But is it actually skill? Absolutely not. Player A did not out-bluff or out-read Player B. And Player B played optimally according to basic strategy; folding pocket Kings pre-flop heads-up would be a mathematical blunder. Player A's massive "skill" gains were entirely engineered by the luck of the deal.

There are many different hands and situations in poker that make it very high variance. In some games of pure skill such as chess, a good player will defeat a beginner almost 100% of the time. However, in order to observe the skill levels between players in poker, it requires **tens of thousands of hands** for a player's true win rate to overcome the noise of random distribution. Nonetheless, it's nice to have these mathematical formulas to quantize a player's luck versus skill level. However, defining skill purely as "equity gained during betting" misses the psychological and macro-strategic elements that define elite poker. Perhaps the greatest skill in poker has nothing to do with the cards at all: it is **game selection**. Recognizing which tables feature weaker

opponents and navigating the mathematical landscape of human error and variance is where the real edge lies.